

Embedded Systems

NXP

Summer school

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Course Contents

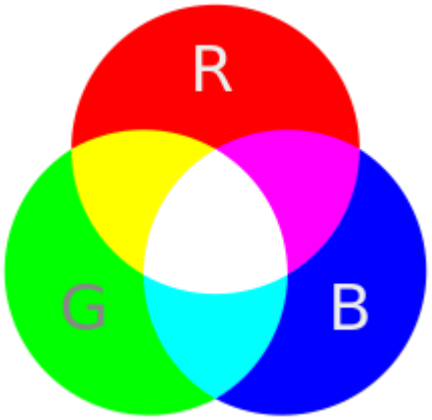
General topics, **Courses** & **Labs**

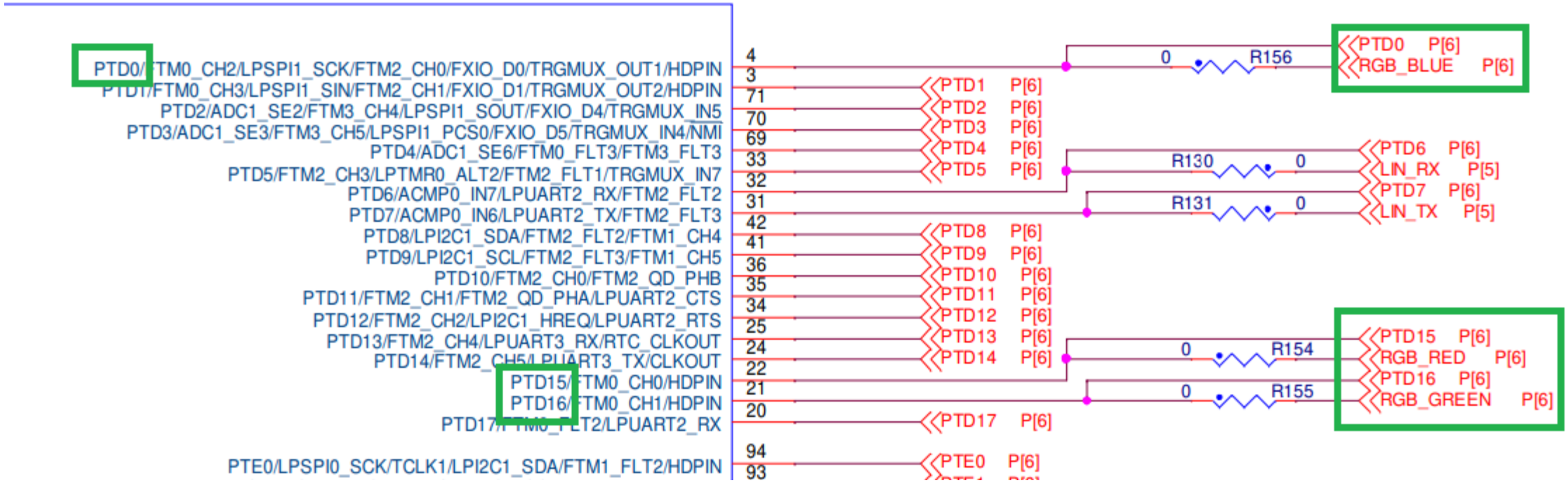
1. General Presentation of the Course
2. V-Model, Requirements Engineering, Process | Understand and Create Requirements
3. Process | Git, IDE Setup, Compile and Flash the Hello World Project
4. Architecture (UML) | Virtual Machine Environment Setup
5. RGB | Hands-on Lab (no module – just the dev board)
6. Node 1: Brake
7. Node 2: Lights
8. Review and Exercises

Exercise: RED-YELLOW-BLUE



- The initial demo project will make the RGB LED to blink rapidly RED and GREEN colors
- Scope of this exercise is to make the RGB LED to blink on RED, YELLOW and BLUE

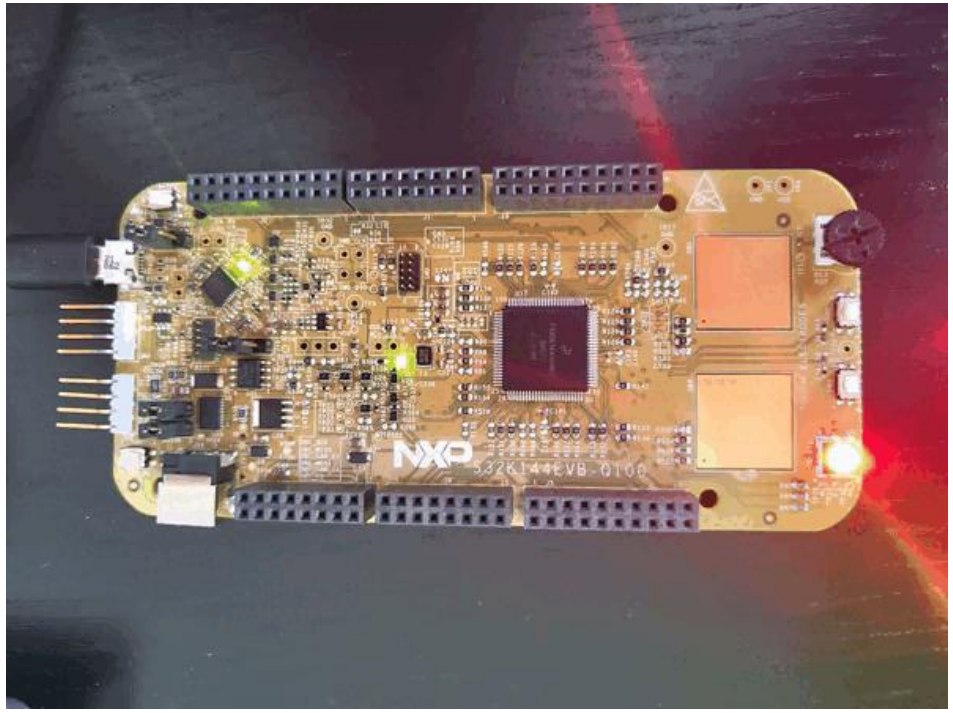




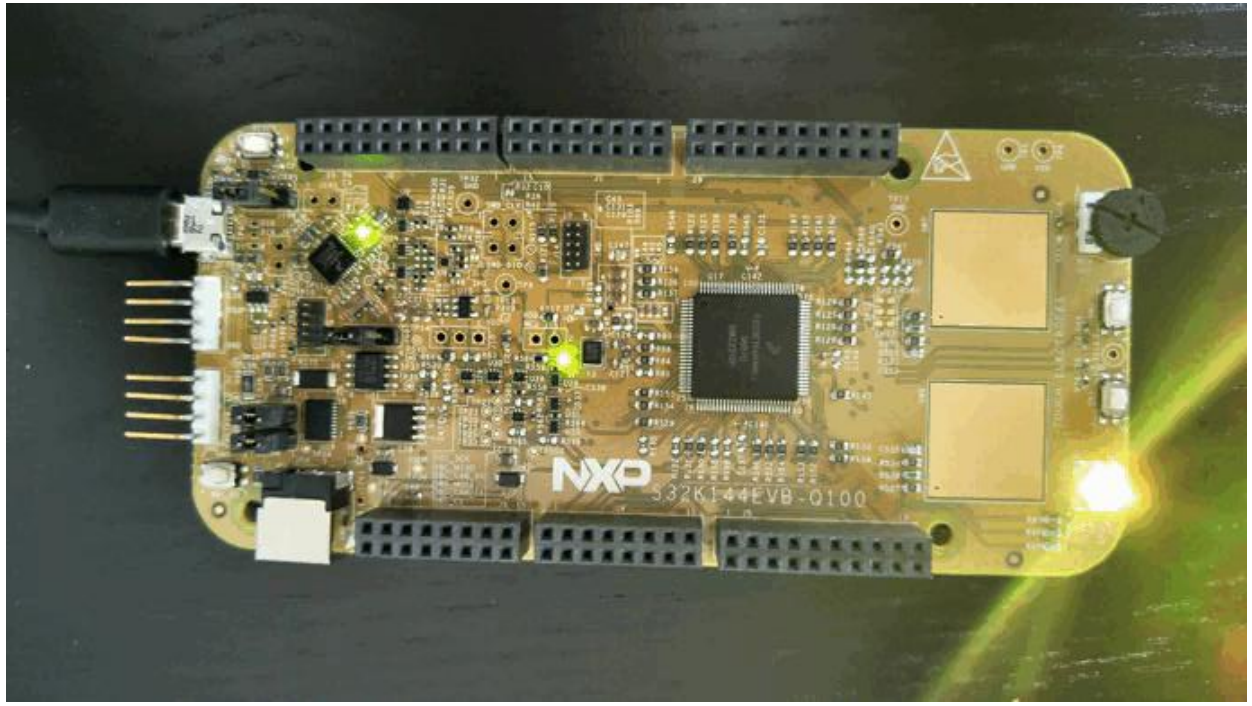
- RGB_BLUE – PTD0 ; RGB_RED – PTD15 ; RGB_GREEN – PTD16

Board pinout

Exercise: RED-YELLOW-BLUE

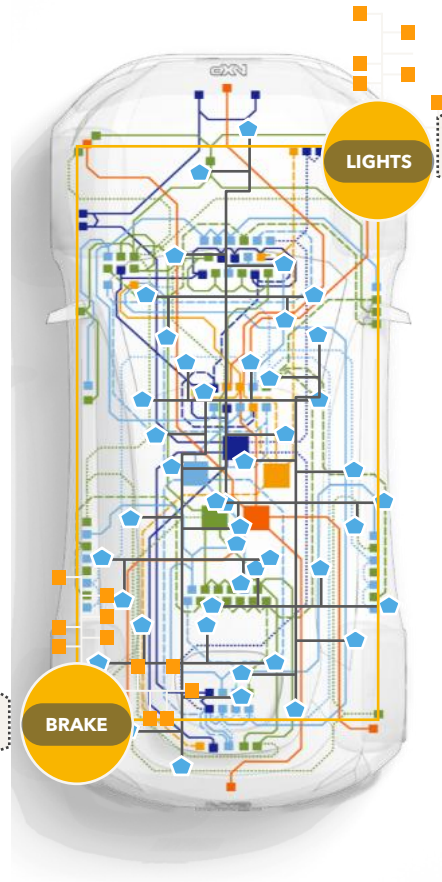


Before



Target result

REPLI-CAR NETWORK



LIGHTS CONTROL NODE
Replicate Body Control Module behavior
· Actuate lights (headlamps, turn indicators), horn and washer pump based on input controller from HMI
· Communications with other nodes: HMI, PCM, Brake, Steering, Comfort

BRAKE CONTROL NODE
Replicate Brake Control Module behavior
· brake actuation and brake lamp indication based on input controller from HMI
· Communications with other nodes: HMI, PCM, BCM

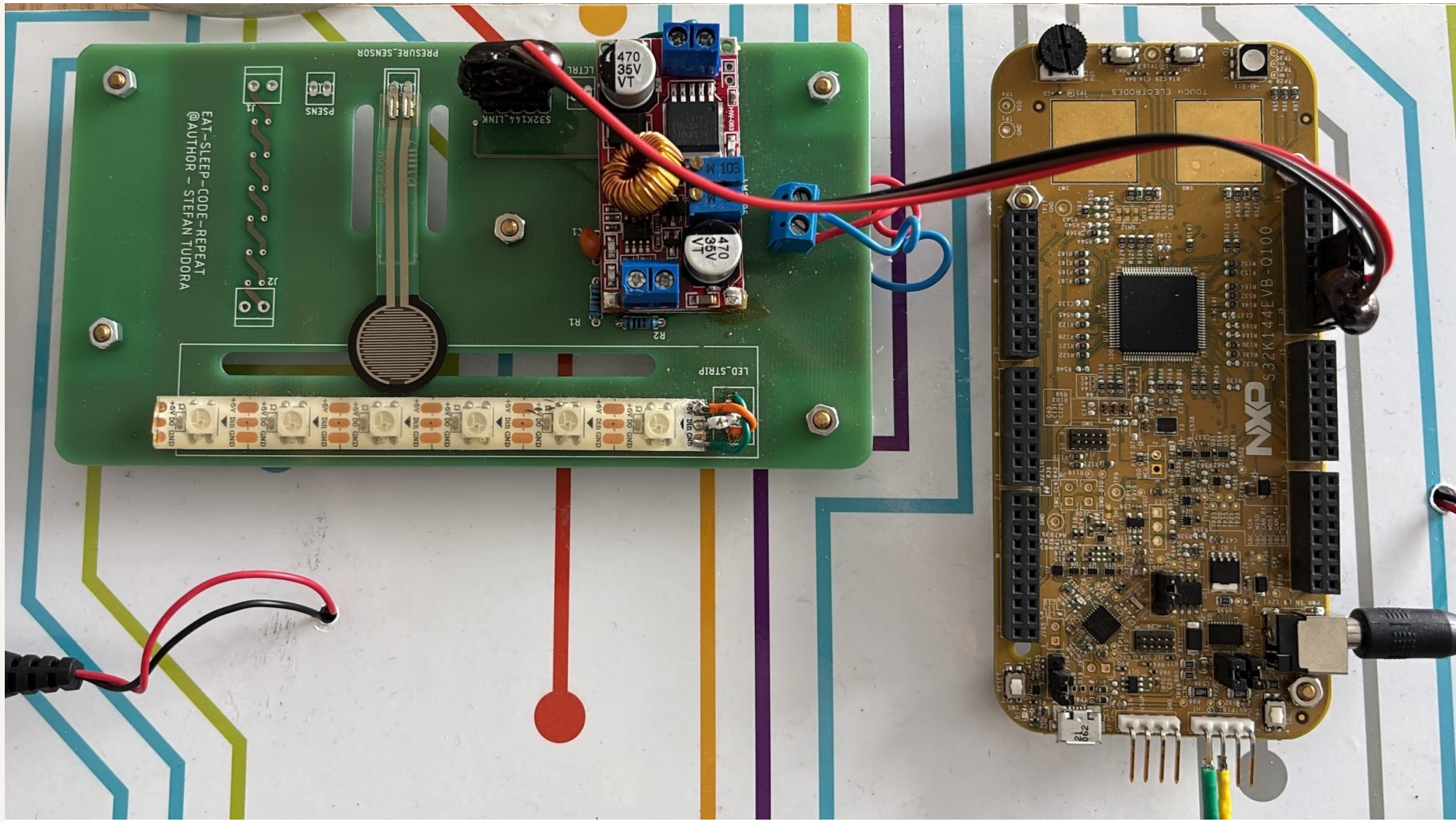
BRAKES



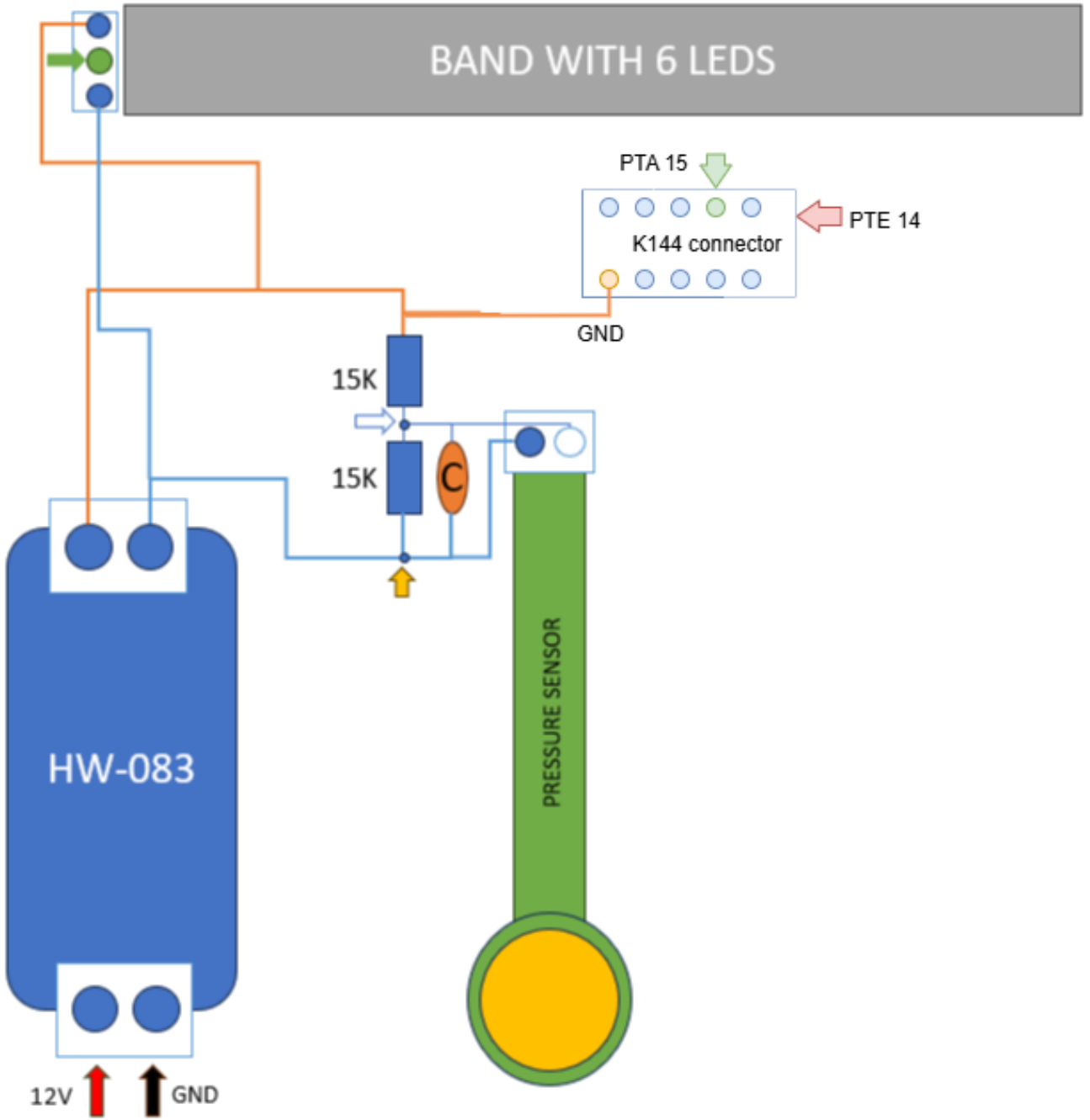
Exercise: BRAKES

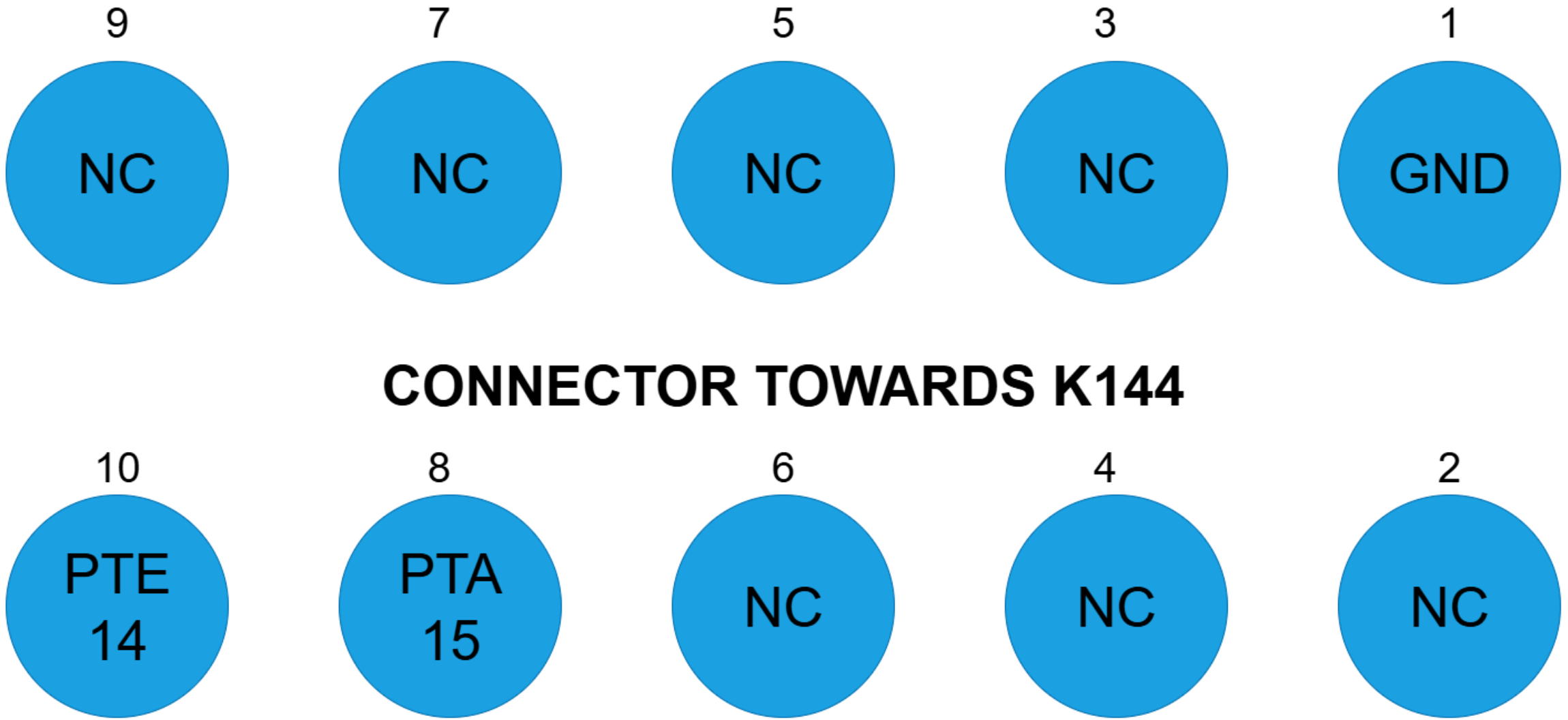


BRAKES


PRESURE_SENSOR
EAT-SLEEP-CODE-REPEAT
@AUTHOR - STEFAN TUDORA
PSENS
LED_STRIP
470
35V
VT
NXP
S32K144EVB-Q100

HW Brakes Module Diagram




9
NC
7
NC
5
NC
3
NC
1
GND
CONNECTOR TOWARDS K144
10
PTE 14
8
PTA 15
6
NC
4
NC
2
NC

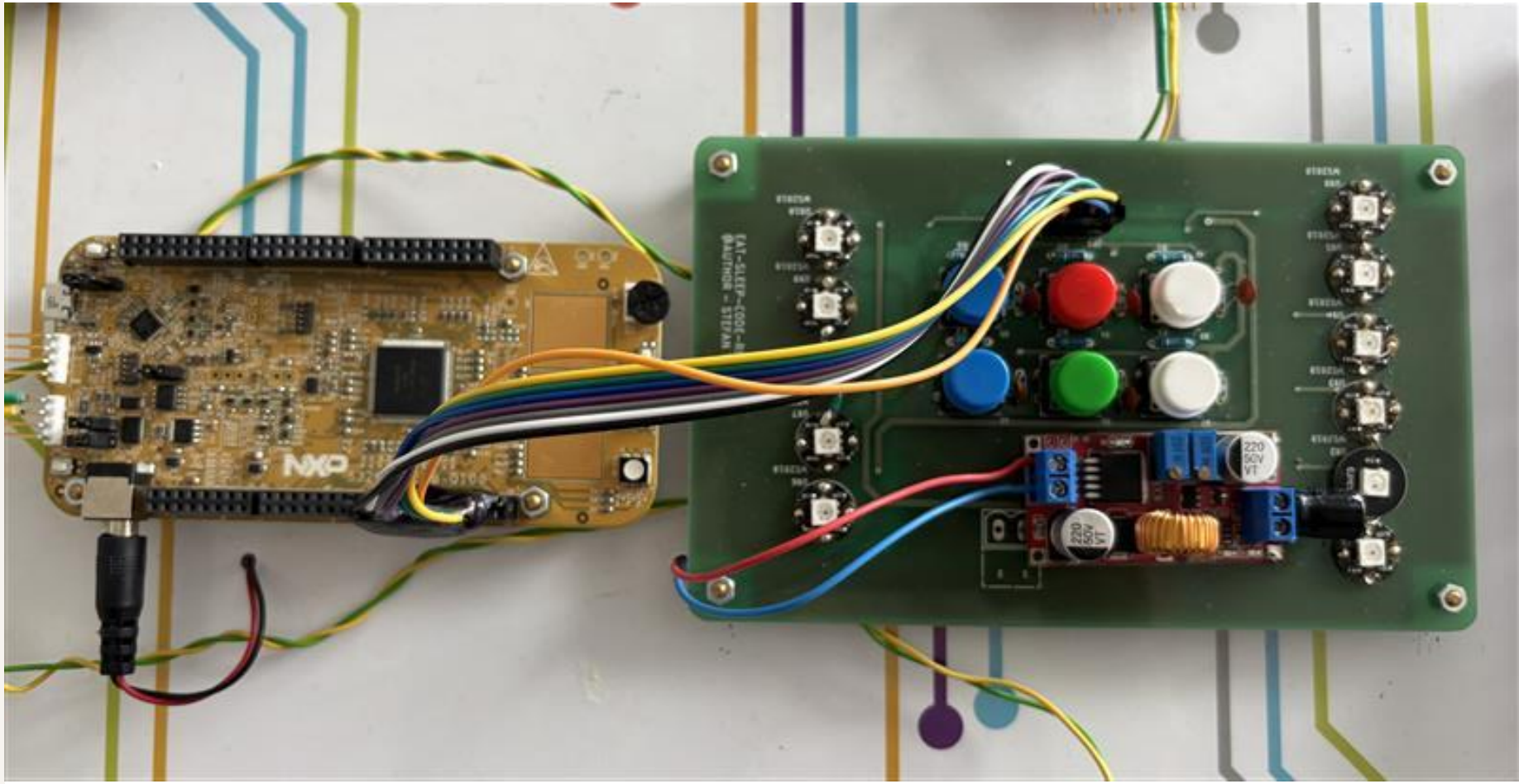
LIGHTS



Exercise: LIGHTS

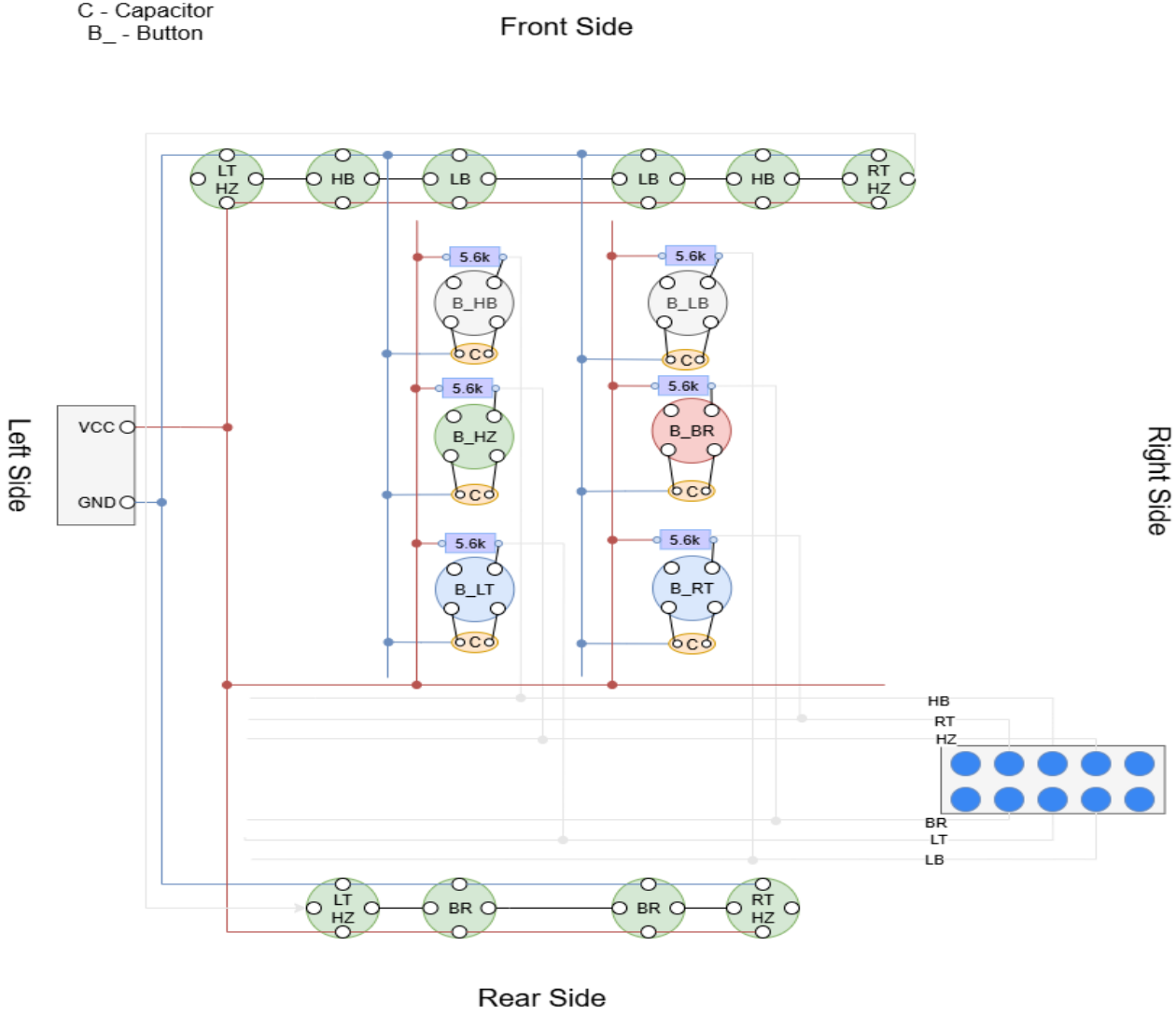


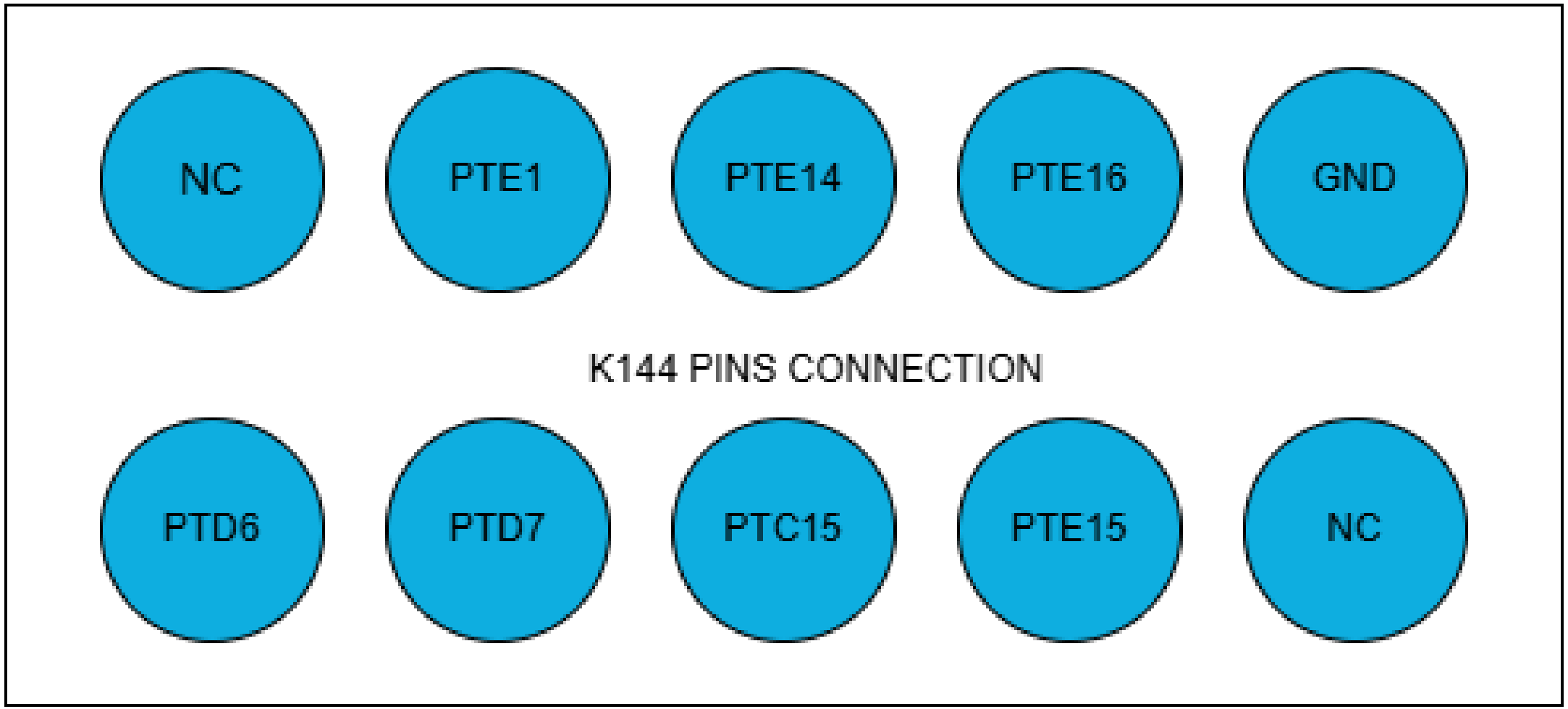
LIGHTS



HW Module Diagram

LB - Low Beam
HB - High Beam
BR - Brake Lights
HZ - Hazard Light
LT - Left Turn
RT - RIght Turn
C - Capacitor
B_ - Button




NC
PTE1
PTE14
PTE16
GND
K144 PINS CONNECTION
PTD6
PTD7
PTC15
PTE15
NC



Brighter
Together

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